

SymPy Bug Report

1 Bug 26: solveset returns a false solution for $\text{acot}(x) = -\pi/2$

1.1 Minimal Reproducer

```
from sympy import Eq, S, acot, pi, simplify, solveset
from sympy.abc import x

sol = solveset(Eq(acot(x), -pi/2), x, S.Reals)
print("solution:", sol)
print("residual at 0:", simplify(acot(0) + pi/2))
print("acot(0):", acot(0))
```

1.2 Observed Behavior

```
solution: {}
residual at 0: pi
acot(0): pi/2
```

SymPy returns 0 as a real solution of $\text{acot}(x) = -\pi/2$. Substitution immediately contradicts that result: SymPy evaluates $\text{acot}(0)$ as $\pi/2$, so the residual for the equation $\text{acot}(x) + \pi/2 = 0$ is π , not 0.

1.3 Expected Behavior

The correct real solution set is $\emptyset$. In SymPy notation, the expected result is `EmptySet`.

1.4 Mathematical Explanation

The equation being solved is

$$\text{acot}(x) = -\frac{\pi}{2}.$$

For SymPy's principal real branch, $\text{acot}(0) = \pi/2$, and the diagnosis records the relevant branch range as $(-\pi/2, \pi/2]$. The left endpoint $-\pi/2$ is not included in that range. Therefore no real x has principal inverse cotangent exactly equal to $-\pi/2$.

The returned value $x = 0$ is not an alternative branch representation or a harmless simplification. It gives

$$\text{acot}(0) - \left(-\frac{\pi}{2}\right) = \frac{\pi}{2} + \frac{\pi}{2} = \pi,$$

so it does not satisfy the original equation.

1.5 Source-Level Diagnosis

The source-level diagnosis locates the problem in two files:

```
sympy/solvers/solveset.py
sympy/functions/elementary/trigonometric.py
```

The relevant code is the generic inverse-function branch of `_invert_real` around lines 230–238, together with `acot.inverse()` around lines 2992–2996.

The traced call path is that `solveset` rewrites the equation as $\text{acot}(x) + \pi/2 = 0$, calls `_solveset`, and then reaches `_invert(acot(x) + pi/2, 0, x, S.Reals)`. The real inversion path subtracts $\pi/2$, reducing the target equation to $\text{acot}(x) = -\pi/2$, and then handles $\text{acot}(x)$ through the generic inverse-function branch.

The diagnosis identifies the following relevant rule:

```
if hasattr(f, 'inverse') and f.inverse() is not None and not isinstance(f, (
    TrigonometricFunction,
    HyperbolicFunction,
)):
    ...
    return _invert_real(f.args[0],
                        imageset(Lambda(n, f.inverse()(n)), g_ys),
                        symbol)
```

For $\text{acot}(x) = -\pi/2$, this branch applies `acot.inverse()`, namely `cot`, to the target value and obtains $\cot(-\pi/2) = 0$. The missing step is a range check for the principal real branch of `acot`. Since $-\pi/2$ is outside the documented principal range $(-\pi/2, \pi/2]$, applying `cot` here is not a valid two-sided inverse step. A plausible fix direction, supported by the diagnosis, is to make this inversion branch range-aware for inverse trigonometric functions, either by intersecting the target set with the principal range before inversion or by special-casing inverse trigonometric functions with branch-aware checks.

1.6 Additional Failing Instances

The independent verification checks the same endpoint error after translating the input by a real shift a :

$$\text{acot}(x - a) = -\frac{\pi}{2}.$$

In each case, SymPy returns a set containing the witness $x = a$. That witness makes the inner argument $x - a$ equal to 0, so SymPy’s own evaluation gives $\text{acot}(0) = \pi/2$ and the same residual π . Thus the expected real solution set is empty across the family.

Input / Expression	SymPy behavior	Expected behavior
<code>solveset(acot(x) = -$\pi/2$, x, $\mathbb{R}$)</code>	returns <code>{0}</code> ; residual at 0 is π	$\emptyset$
<code>solveset(acot(x - 1) = -$\pi/2$, x, $\mathbb{R}$)</code>	contains 1; residual at 1 is π	$\emptyset$
<code>solveset(acot(x + 1) = -$\pi/2$, x, $\mathbb{R}$)</code>	contains -1; residual at -1 is π	$\emptyset$
<code>solveset(acot(x - 2) = -$\pi/2$, x, $\mathbb{R}$)</code>	contains 2; residual at 2 is π	$\emptyset$
<code>solveset(acot(x + 3) = -$\pi/2$, x, $\mathbb{R}$)</code>	contains -3; residual at -3 is π	$\emptyset$
<code>solveset(acot(x - π) = -$\pi/2$, x, $\mathbb{R}$)</code>	contains π ; residual at π is π	$\emptyset$

1.7 Related Issues Assessment

The bug-finding harness performed a best-effort deduplication check. The closest public match identified was SymPy issue #17334, a broad discussion of trigonometric solving and inversion limitations in `solveset`; however, the available evidence did not find a public report for `acot(x) = - $\pi/2$` , for `solveset` returning `{0}`, or for this exact `acot(0)` residual failure. The deduplication verdict is therefore that this is a new instance of a known failure mode: inverse-trigonometric branch and range mistakes are a known family, while this exact reproducible endpoint case is previously unreported and remains individually actionable as an issue and regression test.

1.8 Proposed SymPy Regression Test

```
import pytest

from sympy import Eq, S, acot, pi, simplify, solveset
from sympy.abc import x

@pytest.mark.parametrize("shift", [0, 1, -1, 2, -3, pi])
def test_acot_endpoint_is_not_inverted_to_false_solution(shift):
    sol = solveset(Eq(acot(x - shift), -pi/2), x, S.Reals)
    assert sol.contains(shift) is not S.true
    assert simplify(acot(shift - shift) + pi/2) != 0
```